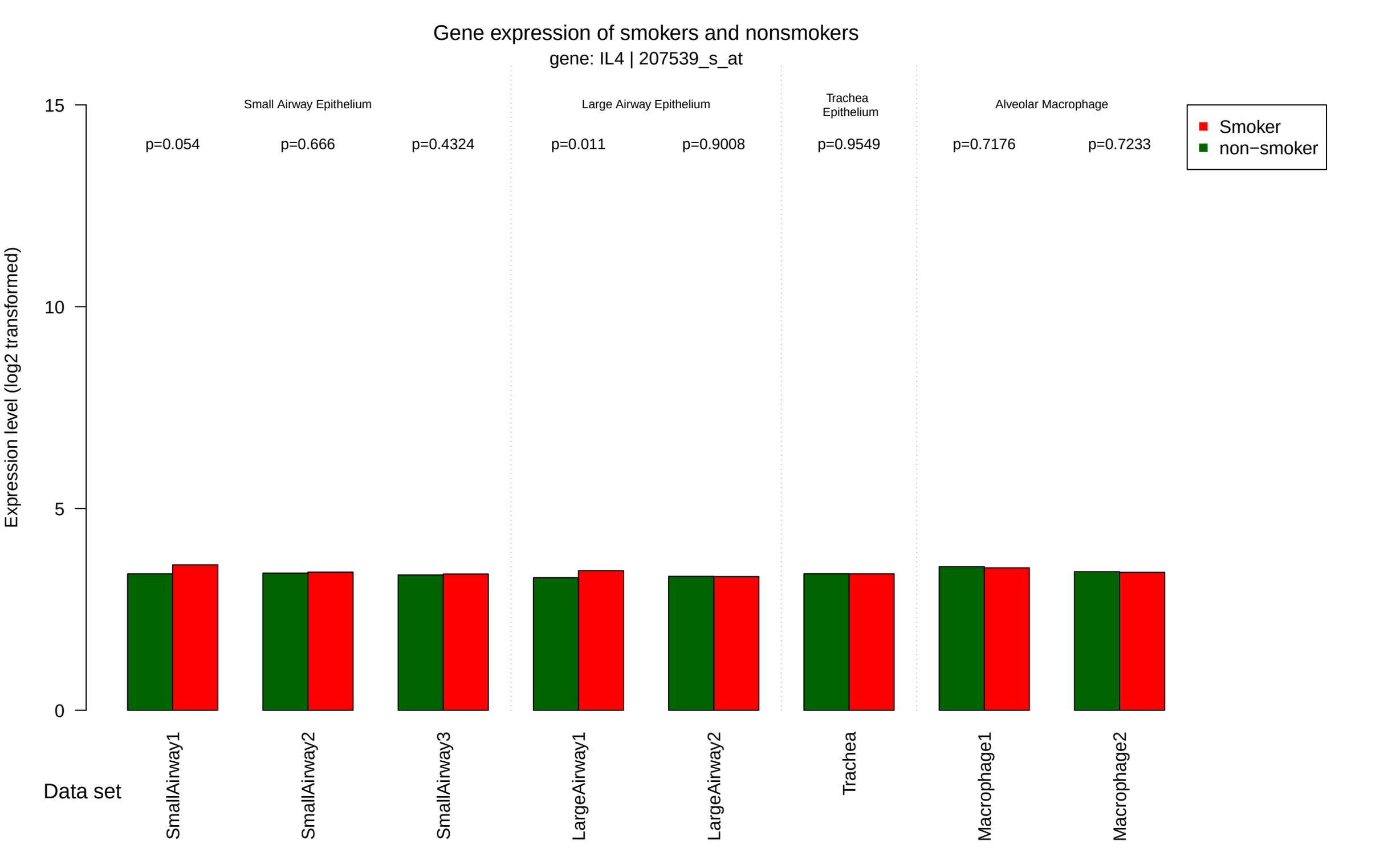
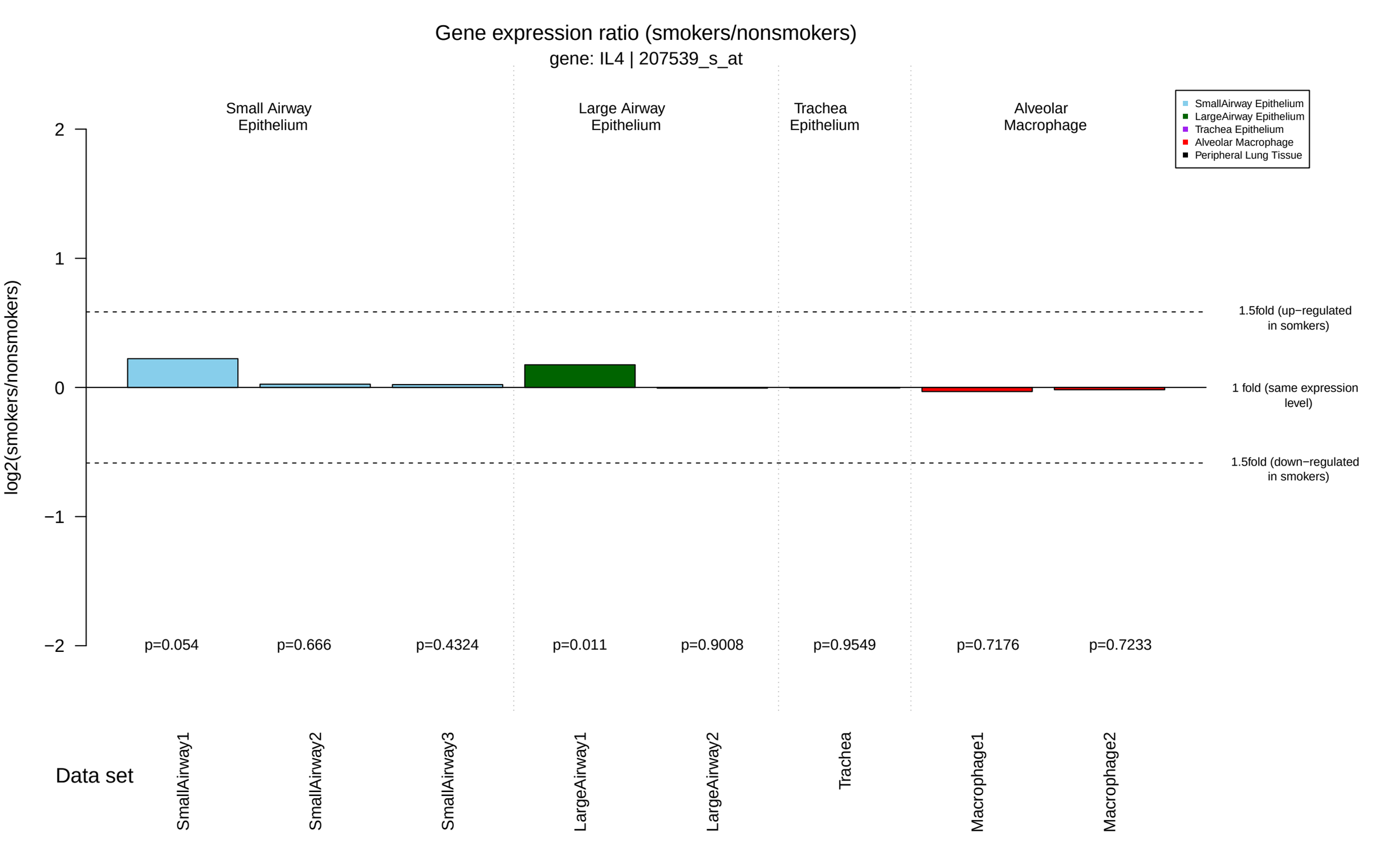
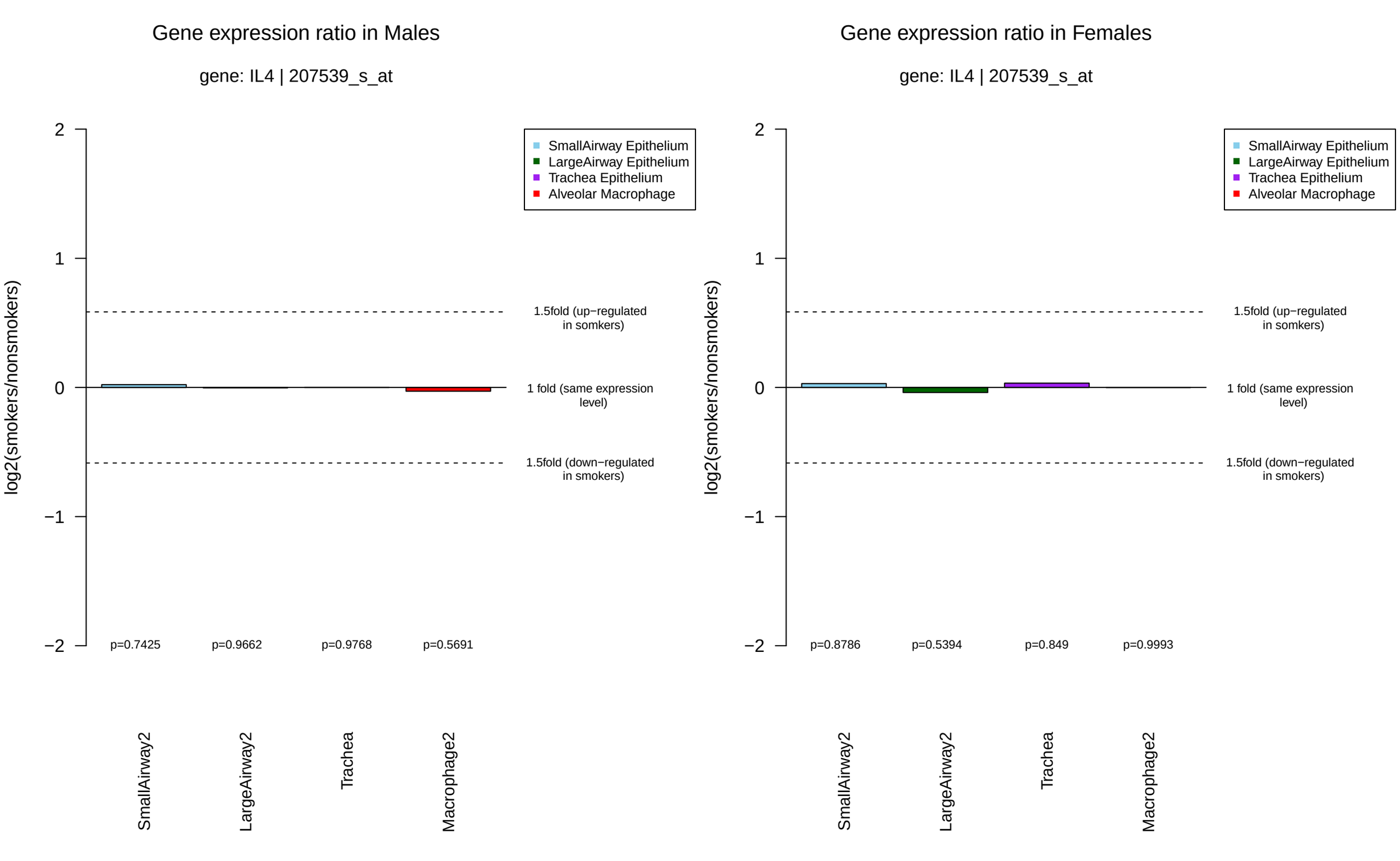
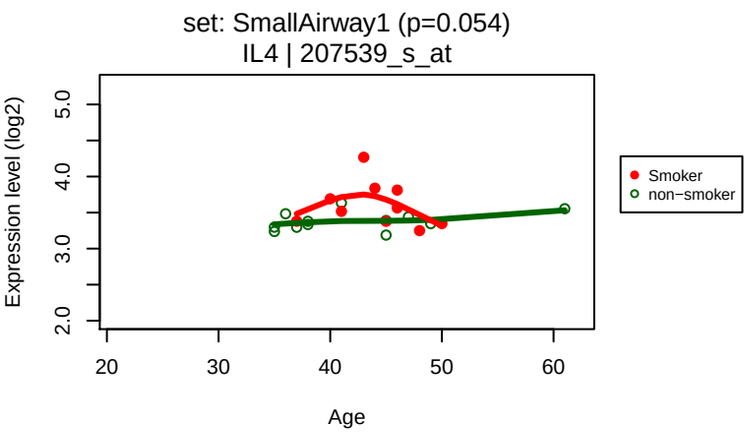
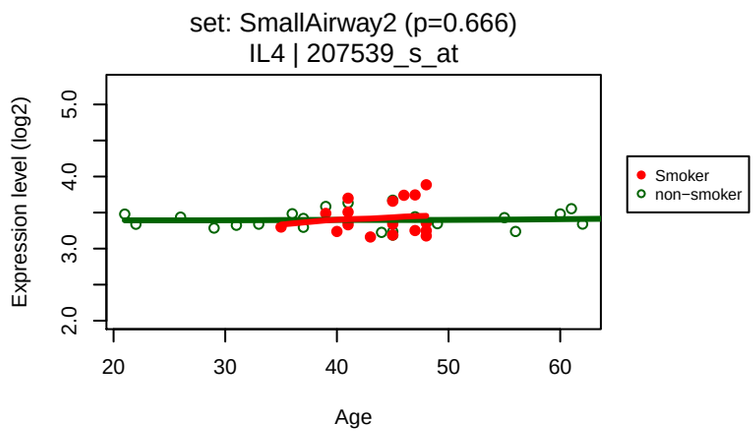
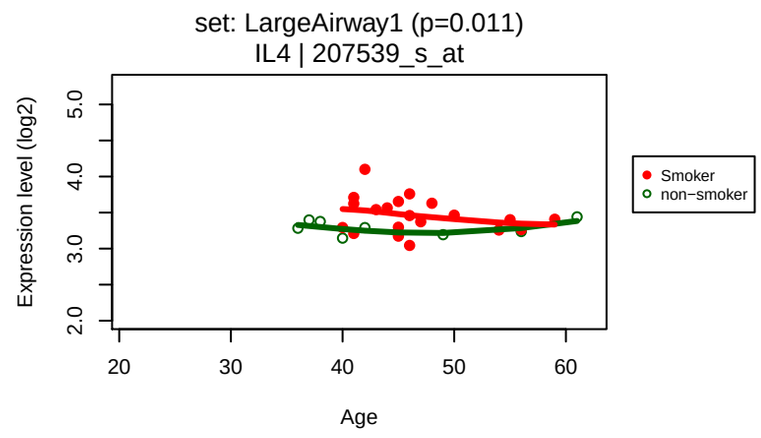

Gene expression of smokers and nonsmokers
gene: IL4 | 207539_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.054
p=0.666
p=0.4324
p=0.011
p=0.9008
p=0.9549
p=0.7176
p=0.7233
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

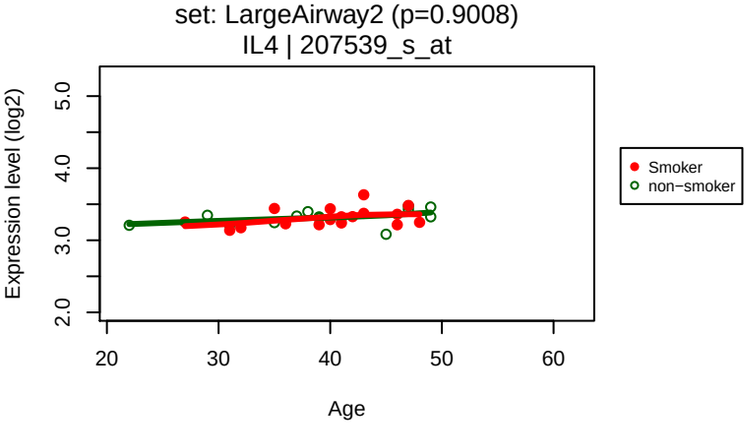

Gene expression ratio (smokers/nonsmokers)
gene: IL4 | 207539_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.054
p=0.666
p=0.4324
p=0.011
p=0.9008
p=0.9549
p=0.7176
p=0.7233
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

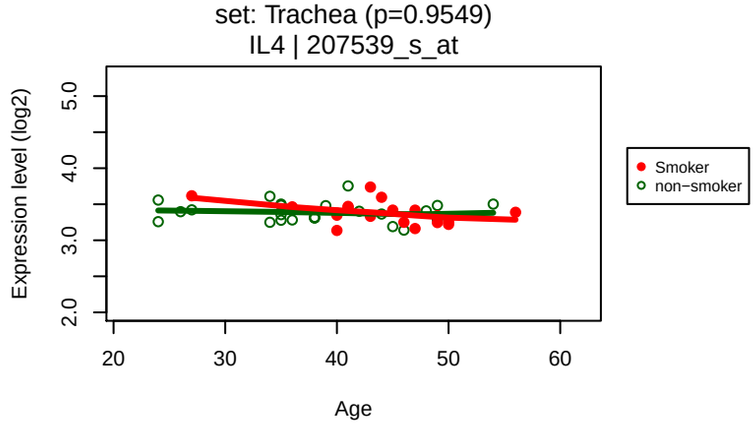

Gene expression ratio in Males
gene: IL4 | 207539_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7425
p=0.9662
p=0.9768
p=0.5691
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL4 | 207539_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8786
p=0.5394
p=0.849
p=0.9993
SmallAirway2
LargeAirway2
Trachea
Macrophage2

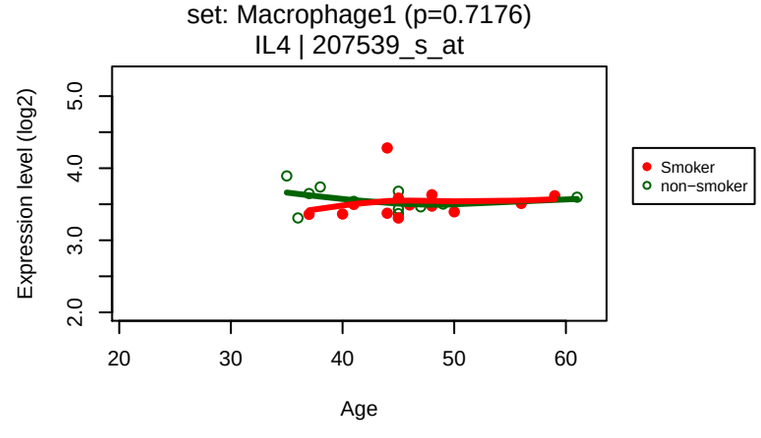

set: SmallAirway1 (p=0.054)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker

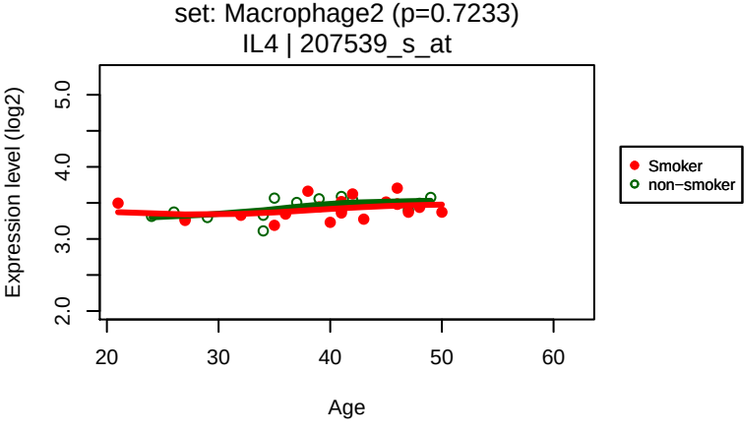

set: SmallAirway2 (p=0.666)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker

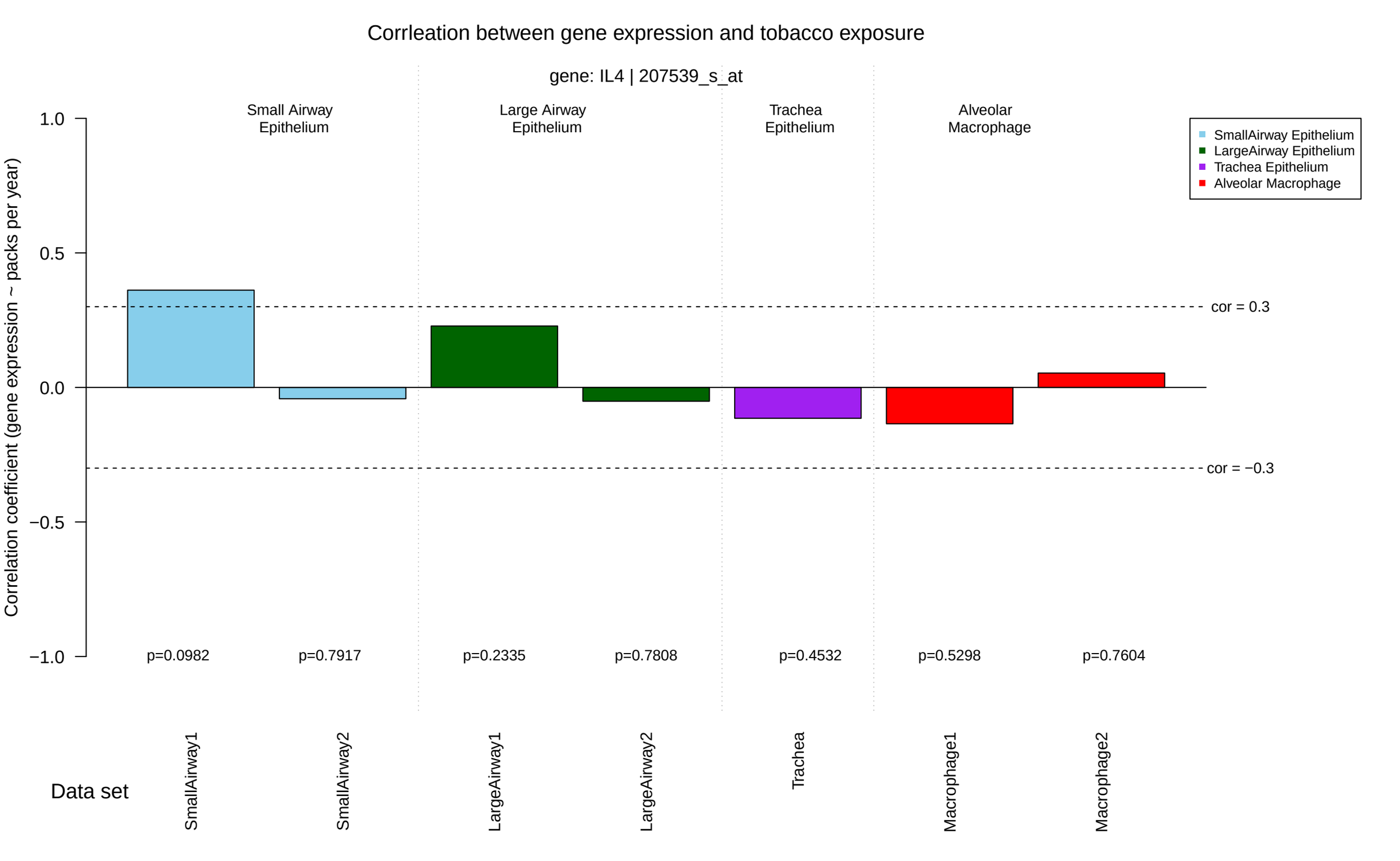

set: LargeAirway1 (p=0.011)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9008)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9549)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7176)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7233)
IL4 | 207539_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL4 | 207539_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0982
p=0.7917
p=0.2335
p=0.7808
p=0.4532
p=0.5298
p=0.7604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2